BIO 004 Human Anatomy · Module 4 notes · Dr. Sharilyn Rennie

The digestive system, part III: small intestine, large intestine, and rectum

The written notes for the third digestive lecture: the small intestine and its three levels of folding, the large intestine, the rectum and anal canal, the whole path of food, and the canal disorders. Module 4 competencies 116 to 119, with nine worked questions.

To print: these notes as a two-column PDF (opens in a new tab). Already printed the Module 4 notes packet? These are the lecture notes, a different document, so print these too if you want them on paper.

How to use these notes. Draw the note sheet for each competency first, from memory, before you read. Then read the matching section here to fill in anything you missed, and use it to review. Every section carries its competency number, so you can go from the competency packet to the page and back. The worked questions at the end show how a lecture exam asks about this material and why each wrong answer is wrong.

Competency 116. Small intestine and surface area

In your course notes: Module 4 packet, Chapter 3, competency 116 (page 33).

Dr. Rennie's video for this part: Digestive system concept video, chapter Small and large intestines.

You should be able to: Identify the duodenum, jejunum, and ileum and name the circular folds, villi, lacteal, and microvilli as features that expand absorptive surface area.

The small intestine runs from the pyloric sphincter to the ileocecal valve. It is the longest part of the canal, about 3 m (10 feet) in a living person, and fills most of the middle and lower abdomen. Most chemical digestion and nearly all absorption of nutrients happen here.

Region	Length	Location and features
Duodenum	About 25 cm (10 inches); the shortest	A C-shaped curve from the pyloric sphincter around the head of the pancreas. Mostly retroperitoneal. Bile and pancreatic juice enter through the major duodenal papilla on its inner wall
Jejunum	About 1 m (3 feet)	The middle region, mostly in the upper left abdomen, hanging on the mesentery. The name means empty, which is how it is found after death
Ileum	About 2 m (6 feet); the longest	The last region, mostly in the lower right abdomen, hanging on the mesentery. Ends at the ileocecal valve. Its wall holds Peyer patches

- There is no sharp border between the jejunum and the ileum; one becomes the other along the mesentery.
- The ileum is the longest part, not the jejunum. Your slides give the duodenum as 12 inches; OpenStax and Tortora give about 25 cm (10 inches). Either is accepted.

The wall of the small intestine

Layer	In the small intestine
Mucosa	Simple columnar epithelium with several cell types: absorptive cells (enzymes on their surface, microvilli for absorption), goblet cells (mucus), and in the intestinal glands (crypts of Lieberkuhn) Paneth cells (lysozyme, a bactericidal enzyme, and phagocytosis) and enteroendocrine cells: S cells (secretin), CCK cells (cholecystokinin), and K cells (glucose-dependent insulinotropic peptide). Lamina propria of areolar connective tissue with MALT: solitary lymphatic nodules, most numerous in the distal ileum, and aggregated lymphatic follicles (Peyer patches) in the ileum. Muscularis mucosae of smooth muscle
Submucosa	In the duodenum, duodenal (Brunner) glands that secrete alkaline mucus to neutralize the acid in chyme. Lymphatic tissue sometimes extends down from the lamina propria
Muscularis	Two layers: a thicker inner circular layer and a thinner outer longitudinal layer
Serosa	Visceral peritoneum completely surrounds the small intestine, except the retroperitoneal part of the duodenum

Three levels of folding, largest to smallest

Feature	What it is	Seen with
Circular folds (plicae circulares)	Permanent ridges of mucosa and submucosa running around the inside of the tube, from the proximal duodenum to the middle of the ileum. Some go all the way around, some part way. Unlike rugae they do not flatten when the tube fills. They increase the surface and make chyme spiral instead of running straight through	The naked eye
Villi	Finger-like projections of the mucosa, 20 to 40 per square millimeter, that give the lining a velvety look. Each has a core of lamina propria with an arteriole, a venule, a capillary network, and a lacteal	A light microscope
Microvilli	Tiny projections of the apical membrane of each absorptive cell, each a bundle of 20 to 30 actin filaments, about 200 million per square millimeter. Together they form the brush border, which also carries digestive enzymes	An electron microscope (a fuzzy edge on a light slide)

- Fold, finger, fuzz. Circular folds are folds of the wall. Villi are fingers on the folds. Microvilli are the fuzz on each cell of each finger.
- The lacteal is the lymphatic capillary in the center of each villus. Most absorbed fats are too large to enter a blood capillary, so they enter the lacteal and ride the lymph before reaching the blood. Sugars and amino acids enter the blood capillaries.

Two hormones from the small intestine

Hormone	Released by	Trigger	Main effects
Cholecystokinin (CCK)	CCK cells of the intestinal glands	Chyme with amino acids and fatty acids	Pancreatic juice rich in enzymes; contraction of the gallbladder, squeezing bile down the cystic and common bile ducts; relaxation of the sphincter of Oddi; slower gastric emptying; a feeling of fullness (satiety) through the hypothalamus; growth of the pancreas; enhances secretin
Secretin	S cells of the intestinal glands	Acidic chyme	Pancreatic juice rich in bicarbonate, which buffers the acid; less gastric acid and gastric juice; growth of the pancreas; enhances CCK

Together they keep the duodenum from being overloaded with more chyme than it can handle. Other gut hormones: motilin, substance P, and bombesin stimulate motility; vasoactive intestinal peptide (VIP) stimulates secretion of ions and water and inhibits gastric acid; gastrin-releasing peptide stimulates gastrin release; somatostatin inhibits it.

Competency 117. Large intestine features

In your course notes: Module 4 packet, Chapter 3, competency 117 (page 34).

Dr. Rennie's video for this part: Digestive system concept video, chapter Small and large intestines.

You should be able to: Identify the cecum, appendix, four parts of the colon, rectum, and anal canal along with the teniae coli, haustra, and omental appendices.

The large intestine is the last part of the canal, from the ileocecal valve to the anus, about 1.5 m (5 feet) long and about 6.5 cm (2.5 inches) across, the same in the living person and the cadaver. It frames the small intestine on three sides. It absorbs water, ions, and some vitamins, forms and stores feces, and empties them. Bacteria living in it break down leftover proteins and make some B vitamins and vitamin K. Haustral churning, peristalsis, and mass peristalsis drive the contents toward the rectum.

The regions, in order

Region	Where it is
Cecum	A blind pouch in the lower right abdomen hanging below the ileocecal valve. The first part of the large intestine
Vermiform appendix	A narrow, worm-shaped tube attached to the cecum, full of lymphatic tissue. Its own small mesentery, the mesoappendix, ties it to the ileum. Food never travels through it
Ascending colon	Up the right side to the liver. Retroperitoneal
Hepatic (right colic) flexure	The bend under the liver where the ascending colon turns left
Transverse colon	Across the upper abdomen from right to left, below the stomach. Hangs on the transverse mesocolon
Splenic (left colic) flexure	The bend near the spleen where the transverse colon turns down
Descending colon	Down the left side. Retroperitoneal
Sigmoid colon	The S-shaped part beginning near the left iliac crest, curving medially into the pelvis to join the rectum. Hangs on the sigmoid mesocolon
Rectum	The straight part in the pelvis, in front of the sacrum and coccyx, from the rectosigmoid junction to the rectoanal junction. Stores feces temporarily; its lower widened part, the ampulla, holds them until defecation. Three transverse rectal folds, one on the right and two on the left
Anal canal	The last 2 to 3 cm, passing through the pelvic floor to open at the anus. Its mucosa is in longitudinal folds, the anal columns, with arteries and veins. Lined with stratified squamous epithelium near the anus. Surrounded by the internal anal sphincter (smooth, involuntary) and the external anal sphincter (skeletal, voluntary), which keep the anus closed except during defecation

- The colon has four parts: ascending, transverse, descending, sigmoid. The two flexures are bends between them, not separate parts.
- Trace it as a frame: up the right side, a bend at the liver, across the top, a bend at the spleen, down the left side, then an S into the pelvis.
- Rectum and anal canal lengths differ between sources, so you are asked where each one sits, not how long it is.

The wall of the large intestine

Layer	In the large intestine
Mucosa	Simple columnar epithelium with absorptive cells (water absorption) and many goblet cells (mucus to lubricate the passage), both in intestinal glands that run the full thickness of the mucosa. No circular folds and no villi, only microvilli on the absorptive cells, so far less absorption than the small intestine. Lamina propria with solitary lymph nodules
Submucosa	Areolar connective tissue
Muscularis	The outer longitudinal layer is gathered into three thick bands, the teniae coli, with little or no longitudinal muscle between them. The inner circular layer is a complete sheet
Serosa	Visceral peritoneum, with the omental appendices hanging from it

Three features found only on the large intestine

Teniae coli

Three flat ribbons of smooth muscle running the length of the cecum and colon: the outer longitudinal layer gathered into bands instead of a full sheet. They spread back into a complete layer at the rectum, so the rectum has no haustra. The three meet at the base of the appendix, which is one way to find it.

Haustra

The row of pouches that gives the colon its puckered look. The teniae are shorter than the wall and keep a tonic contraction, so the wall bunches up between them.

Omental (epiploic) appendices

Small fat-filled pouches of visceral peritoneum hanging from the surface of the colon, mostly along the teniae.

These three are how you tell large from small intestine at a glance on a model or the cadaver. Count the teniae: three, not two or four.

Blood and nerves of the rectum

- Arteries: the superior rectal artery (from the inferior mesenteric artery) supplies the upper rectum; the right and left middle rectal arteries supply the lower rectum.
- Veins: a venous plexus in the wall drained by the rectal veins.
- Nerves: sympathetic fibers from the lumbar splanchnic nerves through the hypogastric plexus; parasympathetic and visceral sensory fibers through the pelvic splanchnic nerves.

Competency 118. Trace the pathway of food

In your course notes: Module 4 packet, Chapter 3, competency 118 (page 35).

You should be able to: Trace a bite of food through every named segment of the alimentary canal in order from mouth to anus.

A bite of food moves through the alimentary canal in one direction only. Every segment below is one the food passes through, in order, with the sphincters at the borders where they sit.

1. Oral cavity: food comes in through the lips, is chewed by the teeth, mixed with saliva, and shaped into a bolus by the tongue.
2. Fauces: the tongue pushes the bolus back through the opening into the throat.
3. Oropharynx: the soft palate and uvula rise to seal the nasopharynx above.
4. Laryngopharynx: the epiglottis covers the larynx, so the bolus passes behind it toward the esophagus.
5. Upper esophageal sphincter relaxes.
6. Esophagus: waves of muscle contraction carry the bolus down through the thorax and the esophageal hiatus.
7. Lower esophageal sphincter relaxes.
8. Cardia of the stomach, then the body, where the food is churned with gastric juice into chyme.
9. Pyloric antrum, pyloric canal, pylorus: the funnel out of the stomach.
10. Pyloric sphincter lets small amounts of chyme through.

11. Duodenum: bile and pancreatic juice enter at the major duodenal papilla.
12. Jejunum: digestion continues and most nutrients are absorbed.
13. Ileum: absorption continues.
14. Ileocecal sphincter (valve) lets the remaining material into the large intestine.
15. Cecum, then ascending colon, hepatic flexure, transverse colon, splenic flexure, descending colon, sigmoid colon.
16. Rectum: feces are stored.
17. Anal canal, through the internal and external anal sphincters, to the anus.

- Three named structures are off the path. The nasopharynx carries only air and is sealed during a swallow. The fundus bulges above the cardia, so food can collect there without passing through it. The appendix is a blind tube off the cecum.
- The cecum comes right after the ileum. Do not jump from the ileum to the ascending colon.
- Group the path as the six organs first (mouth, pharynx, esophagus, stomach, small intestine, large intestine), then fill in the named parts inside each one and put a sphincter at each border.
- Most absorption happens in the small intestine. Water is absorbed and feces form in the large intestine.

Competency 119. Clinical anatomy of canal disorders

In your course notes: Module 4 packet, Chapter 3, competency 119 (page 36).

You should be able to: Match GERD, hiatal hernia, peptic ulcer, appendicitis, diverticulosis, IBD, hemorrhoids, and colorectal cancer to the canal structure each one affects.

Each disorder starts in a specific structure. If you know where that structure sits and what layers it has, you can predict where the patient feels it and how serious it can become.

Disorder	Structure affected	What changes
Gastroesophageal reflux disease (GERD)	Lower esophageal sphincter and lower esophagus	The sphincter does not close well, so acidic stomach contents flow back up. The stratified squamous lining of the esophagus has no protection against acid and becomes inflamed. Felt as heartburn, a burning behind the sternum, because the esophagus lies in the thorax just behind it
Hiatal hernia	Esophageal hiatus and upper stomach	Part of the stomach pushes up through the hiatus into the thorax. The diaphragm loses its grip on the lower esophageal sphincter, so reflux often follows
Peptic ulcer	Mucosa of the stomach or duodenum	An open sore where the mucosa has worn away. Common causes: infection with Helicobacter pylori and regular use of NSAID pain relievers. It deepens layer by layer: through the submucosa, where it can open a vessel and bleed, through the muscularis, and through the serosa, where it perforates and the contents spill into the peritoneal cavity
Appendicitis	Vermiform appendix	The lumen blocks, the appendix swells and inflames, and pain lands in the lower right abdomen where the cecum sits. If it bursts, its contents spill into the peritoneal cavity: peritonitis
Diverticulosis	Wall of the colon, most often the sigmoid	Small pouches (diverticula) of mucosa and submucosa bulge out through weak spots in the muscularis. When the pouches inflame, it is diverticulitis
Inflammatory bowel disease (IBD)	Wall of the small and large intestine	Long-term inflammation of the wall. Crohn disease can affect any part from mouth to anus, most often the ileum and colon, through every layer. Ulcerative colitis affects only the colon and rectum, mostly the mucosa
Hemorrhoids	Veins in the wall of the anal canal and lower rectum	The veins swell and stretch into soft bulges. External hemorrhoids are thrombosed veins of the external plexus, often with pregnancy or constipation, and painful because they have somatic innervation. Internal hemorrhoids are dilated veins of the internal rectal plexus above the dentate line, visceral innervation, so painless unless they ulcerate
Colorectal cancer	Mucosa of the colon or rectum	Lining cells grow out of control, usually starting as a polyp, then growing through the deeper layers of the wall

What this means for patients

- Where a problem sits predicts the symptom. A weak lower esophageal sphincter burns behind the sternum. An inflamed appendix hurts in the lower right.
- The four-layer wall explains how a problem gets worse. Ulcers, diverticula, IBD, and cancer all become more serious as they reach deeper layers, and a perforation through the serosa turns a gut problem into peritonitis.
- GERD and hiatal hernia are two different things that often travel together: one is a weak sphincter, the other is stomach through the diaphragm.

Worked questions

Nine questions in the style of the lecture exam. Read the answer and the reasons each wrong option is wrong. The wrong options are where the learning is.

QUESTION 1 · COMPETENCY 116

Which region of the small intestine is the longest?

A. Duodenum

B. Jejunum

C. Ileum

D. They are all about the same length

Answer: C. The ileum is about 2 m (6 feet), the jejunum about 1 m (3 feet), and the duodenum about 25 cm (10 inches). The ileum is the last and longest region.

Why the others are wrong

- **A** The duodenum is the shortest, a C around the head of the pancreas.
- **B** The jejunum is the middle region and about half the length of the ileum.
- **D** The three differ a lot; the duodenum is a fraction of the length of the other two.

QUESTION 2 · COMPETENCY 116

The lacteal in the center of a villus is

A. a blood capillary that absorbs sugars

B. a lymphatic capillary that absorbs most fats

C. a duct that carries bile into the villus

D. a nerve of the submucosal plexus

Answer: B. The lacteal is a lymphatic capillary. Absorbed fats are too large for blood capillaries, so they enter the lacteal and travel in lymph before reaching the blood.

Why the others are wrong

- **A** The villus does have a blood capillary network, and sugars and amino acids enter it, but that is not the lacteal.
- **C** Bile enters the duodenum at the major duodenal papilla and stays in the lumen. No duct runs into a villus.
- **D** The submucosal plexus sits in the submucosa, below the mucosa. The lacteal is a vessel, not a nerve.

QUESTION 3 · COMPETENCY 116

Put the three surface-expanding features of the small intestine in order from largest to smallest.

A. Villi, circular folds, microvilli

B. Microvilli, villi, circular folds

C. Circular folds, villi, microvilli

D. Circular folds, microvilli, villi

Answer: C. Circular folds are ridges of the whole wall you can see with your eyes. Villi are fingers of mucosa on the folds, seen with a light microscope. Microvilli are the fuzz on one cell, seen with an electron microscope. Fold, finger, fuzz.

Why the others are wrong

- **A** Circular folds are larger than villi; villi sit on the folds.
- **B** This is smallest to largest, the reverse order.
- **D** Microvilli are the smallest, on the surface of a single absorptive cell on a villus.

QUESTION 4 · COMPETENCY 117

Which two parts of the colon hang on a mesocolon?

A. Ascending and descending

B. Ascending and transverse

C. Transverse and sigmoid

D. Descending and sigmoid

Answer: C. The transverse colon hangs on the transverse mesocolon and the sigmoid colon on the sigmoid mesocolon. Those are the two intraperitoneal parts.

Why the others are wrong

- **A** Ascending and descending are retroperitoneal, fixed to the back wall.
- **B** The transverse colon is right, but the ascending colon is retroperitoneal.
- **D** The sigmoid colon is right, but the descending colon is retroperitoneal.

QUESTION 5 · COMPETENCY 117

The teniae coli are

A. three bands of the outer longitudinal muscle layer

B. three bands of the inner circular muscle layer

C. fat-filled pouches of peritoneum on the colon

D. the pouches that give the colon its puckered look

Answer: A. The outer longitudinal layer of the muscularis is gathered into three flat ribbons instead of a complete sheet. Because they are shorter than the wall, the wall bunches into haustra between them.

Why the others are wrong

- **B** The inner circular layer stays a complete sheet in the large intestine.
- **C** Those are the omental appendices, which hang along the teniae.
- **D** Those are the haustra, the pouches the teniae create.

QUESTION 6 · COMPETENCY 117

Which sphincter of the anal canal is smooth muscle and not under voluntary control?

A. External anal sphincter

B. Internal anal sphincter

C. Ileocecal sphincter

D. Pyloric sphincter

Answer: B. The internal anal sphincter is a thickening of the circular smooth muscle of the anal canal. It is involuntary. The external sphincter outside it is skeletal and voluntary.

Why the others are wrong

- **A** The external anal sphincter is skeletal muscle, the one you control.
- **C** The ileocecal sphincter is smooth and involuntary, but it sits between the ileum and the cecum, not at the anal canal.
- **D** The pyloric sphincter is at the exit of the stomach.

QUESTION 7 · COMPETENCY 118

A bite of food leaves the ileum. Which segment does it enter next?

A. Ascending colon

B. Cecum

C. Transverse colon

D. Rectum

Answer: B. The ileum opens through the ileocecal valve into the cecum, the pouch at the start of the large intestine. The ascending colon comes after the cecum.

Why the others are wrong

- **A** The ascending colon rises from the cecum; the food reaches it one step later.
- **C** The transverse colon is two steps and one flexure further on.
- **D** The rectum is near the end of the path, after the sigmoid colon.

QUESTION 8 · COMPETENCY 119

A patient has a burning pain behind the sternum after meals because stomach acid is flowing back into the esophagus. Which structure is failing?

A. Pyloric sphincter

B. Upper esophageal sphincter

C. Lower esophageal sphincter

D. Ileocecal valve

Answer: C. This is GERD. The lower esophageal sphincter does not close well, acid comes up onto the stratified squamous lining of the esophagus, and the esophagus sits in the thorax just behind the sternum, so that is where it burns.

Why the others are wrong

- **A** The pyloric sphincter controls flow out of the stomach into the duodenum. It has nothing to do with backflow into the esophagus.
- **B** The upper esophageal sphincter sits at the top of the esophagus. Acid would have to pass the lower sphincter first.
- **D** The ileocecal valve is between the small and large intestine, far from the sternum.

QUESTION 9 · COMPETENCY 119

A peptic ulcer starts in which layer and erodes in which direction?

A. Serosa, inward toward the lumen

B. Muscularis, in both directions

C. Mucosa, outward toward the serosa

D. Submucosa, inward toward the lumen

Answer: C. An ulcer begins on the inside, where acid and H. pylori meet the mucosa. It deepens through the submucosa (where it can bleed), the muscularis, and finally the serosa, where it perforates into the peritoneal cavity.

Why the others are wrong

- **A** The serosa is the outside of the tube; the acid is on the inside. Ulcers do not start from the peritoneal cavity.
- **B** The muscularis is the third layer in; an ulcer reaches it only after eating through the mucosa and submucosa.
- **D** The submucosa is the second layer. The ulcer gets there from the mucosa and continues outward, not inward.